Power Series : Ordinary Points

$$y'' + p(x)y' + q(x)y = f(x)$$

If $p(x)$, $q(x)$ and $f(x)$ are analytic at x_0, then x_0 will be called an ordinary point.

e^x, $\ln x$, $\cos x$, $\sin x$, polynomials

$e^x \cos x$, $e^x \sin x \longrightarrow$ analytic.

multiplication of power series.

Power Series Multiplication:

$$\sum_{n=0}^{\infty} a_n x^n \quad \text{and} \quad \sum_{n=0}^{\infty} b_n x^n$$

$$\left(\sum_{n=0}^{\infty} a_n x^n\right)\left(\sum_{n=0}^{\infty} b_n x^n\right) = a_0 b_0 + (a_0 b_1 + a_1 b_0) x + (- - - -) x^2 + - - - -$$

$$= \sum_{n=0}^{\infty} c_n x^n$$

$$c_n = \sum_{k=0}^{n} a_n b_{n-k}.$$

Determine the first four terms of the power series sol^n about $x_0 = 0$ to

$$y'' + (\cos x)\, y = 0$$

in terms of $\underbrace{a_0 \text{ and } a_1}$.

Sol^n: $y = \sum_{n=0}^{\infty} a_n x^n$; $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots$

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n$$

Now put all these expressions into the equation!

$$\Rightarrow \sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}x^n + \left(1-\frac{x^2}{2}+\frac{x^4}{4!}-\frac{x^6}{6!}+\cdots\right)\left(\sum_{n=0}^{\infty}a_n x^n\right)=0.$$

$$\Rightarrow \sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}x^n + \sum_{n=0}^{\infty}a_n x^n - \frac{1}{2}\sum_{n=0}^{\infty}a_n x^{n+2} + \cdots$$

$$\cdots = 0$$

$$\Rightarrow 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + \sum_{n=4}^{\infty}(n+2)(n+1)a_{n+2}x^n + a_0 + a_1x + a_2x^2 + a_3x^3 + \sum_{n=4}a_nx^n - \frac{1}{2}a_0x^2 - \frac{1}{2}a_1x^3 - \frac{1}{2}\sum_{n=2}^{\infty}a_nx^{n+2} \cdots = 0$$

Now equating the coefficients, we get

a_0, a_1, a_2, a_3

$$\left.\begin{aligned} 2a_2 + a_0 &= 0 \\ 6a_3 + a_1 &= 0 \\ 12a_4 + a_2 - \tfrac{1}{2}a_0 &= 0 \\ 20a_5 + a_3 - \tfrac{1}{2}a_1 &= 0 \end{aligned}\right\}$$

$$a_2 = -\frac{1}{2}a_0 \quad \text{and} \quad a_3 = -\frac{a_1}{6}.$$

Hence the first four terms of the sol^n is:

$$y = a_0 + a_1 x - \frac{1}{2}a_0 x^2 - \frac{a_1}{6}x^3 + \cdots\cdots$$

(Ans)

Similar Problem:-

$$y'' + (\sin x)\,y' + x\,y = 0$$

$$y'' + (\sin x)\,y' + e^{x}\,y = 1 + 2x^{3},$$

EXR.

Legendre Equation:- The equation which often comes in the study of mathematical physics as follow:

$$(1-x^2)y''-2xy'+\nu(\nu+1)y=0$$

is known as Legendre equation. where $\nu\in\mathbb{R}$.

Our aim is to find power series sol^n about $x_0=0$.

We can divide the equ^n by $(1-x^2)$ $x\neq 1,-1$.

$$y''-\frac{2x}{1-x^2}y'+\frac{\nu(\nu+1)}{1-x^2}y=0. \qquad x\in(-1,1)$$

$$\frac{1}{1-x^2}=\frac{1}{1-y}=1+y+y^2+\cdots=1+x^2+x^4+\cdots$$

$$p(x)=-\frac{2x}{1-x^2}=-2x-2x^3-2x^5-\cdots=0-2.(x-0)-2(x-0)^3-2.(x-0)^5\cdots$$

$x=0$ is an ordinary point since

$$p(x) = -\frac{2x}{1-x^2} \quad \text{and} \quad q(x) = \frac{\nu(\nu+1)}{1-x^2}$$

has power series representation at $x=0$.

Now the series sol^n (say) $y = \sum_{n=0}^{\infty} a_n x^n$

$$(1-x^2)y'' - 2xy' + \nu(\nu+1)y = (1-x^2)\sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}x^n$$

$$-2x\sum_{n=0}^{\infty}(n+1)a_{n+1}x^n + \nu(\nu+1)\sum_{n=0}^{\infty}a_n x^n = 0.$$

$$= \sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}x^n - \sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}x^{n+2}$$

$$-2\sum_{n=0}^{\infty}(n+1)a_{n+1}x^{n+1} + \nu(\nu+1)\sum_{n=0}^{\infty}a_n x^n = 0$$

$$\Rightarrow 2a_2 + 6a_3 x + \sum_{n=2}^{\infty} (n+2)(n+1) a_{n+2} x^n$$

$$- \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^{n+2}$$

$$- 2a_1 x - 2 \sum_{n=1}^{\infty} (n+1) a_{n+1} x^{n+1}$$

$$+ \nu(\nu+1) a_0 + \nu(\nu+1) a_1 x$$

$$+ \nu(\nu+1) \sum_{n=2}^{\infty} a_n x^n = 0.$$

$$\Rightarrow 2a_2 + 6a_3 x + \sum_{n=0}^{\infty} (n+4)(n+3) a_{n+4} x^{n+2}$$

$$- \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^{n+2}$$

$$- 2a_1 x - 2 \sum_{n=0}^{\infty} (n+2) a_{n+2} x^{n+2} + \nu(\nu+1) a_0$$

$$+ \nu(\nu+1) a_1 x$$

$$+ \nu(\nu+1) \sum_{n=0}^{\infty} a_{n+2} x^{n+2} = 0.$$

$\Rightarrow 2a_2 + \nu(\nu+1)a_0 = 0 \quad \Rightarrow a_2 = -\dfrac{\nu(\nu+1)a_0}{2}$

$\Rightarrow 6a_3 - 2a_1 + \nu(\nu+1)a_1 = 0$

and $a_{n+4} = \dfrac{(n+2)(n+3) - \nu(\nu+1)}{(n+4)(n+3)} a_{n+2}, \quad n = 0, 1, 2, \ldots$

$(N=n)$ $n=N$

which can also be written as $\boxed{\nu = N}$ $\quad \boxed{\nu = n} \rightarrow a_{n+2} = 0$

$a_3 \leftarrow \boxed{n=1}$ — $a_{n+2} = -\dfrac{(\nu-n)(\nu+n+1)}{(n+1)(n+2)} a_n, \quad n = 0, 1, 2, \ldots$

$a_4 \leftarrow \boxed{n=2}$ $\quad n=0$

$a_6 \leftarrow \boxed{n=4}$

$$\boxed{a_2 = \frac{-\nu(\nu+1)}{1 \cdot 2} a_0}$$

For even values of n,

$$n=0 \Rightarrow a_2 = -\frac{\nu(\nu+1)}{2} a_0$$

$$n=2 \Rightarrow a_4 = -\frac{(\nu-2)(\nu+3)}{3.4} a_2 = \frac{(\nu-2)\nu(\nu+1)(\nu+3)}{4!} a_0$$

$$n=4 \Rightarrow a_6 = -\frac{(\nu-4)(\nu+5)}{5.6} a_4 = -\frac{(\nu-4)(\nu-2)\nu(\nu+1)(\nu+3)(\nu+5)}{6!} a_0$$

Hence, in general

$$a_{2k} = (-1)^k \frac{(\nu-2k+2)(\nu-2k+4)\cdots(\nu-2)\nu(\nu+1)(\nu+3)\cdots(\nu+2k-1)}{(2k)!} a_0$$

$\hookrightarrow$ Coefficients for the even exponents!

and similarly for the odd term,

$$a_{2k+1} = (-1)^k \frac{(\nu-2k+1)\cdots(\nu-3)(\nu-1)(\nu+2)(\nu+4)\cdots(\nu+2k)}{(2k+1)!}\, a_1$$

Consequently for $a_0 \neq 0$ and $a_1 \neq 0$, two linearly independent sol^n's are

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$$\begin{cases} y_1(x) = \displaystyle\sum_{n=0}^{\infty} a_{2n} x^{2n} = a_0\left(1 - \frac{\nu(\nu+1)}{2}x^2 + \frac{(\nu-2)\nu(\nu+1)(\nu+3)}{4!}x^4 + \cdots\right) \\ y_2(x) = \displaystyle\sum_{n=0}^{\infty} a_{2n+1} x^{2n+1} = a_1\left(x - \frac{(\nu-1)(\nu+3)}{3!}x^3 + \cdots\right) \end{cases}$$

$y_2(x) = a_1(x -$

$$y(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots$$

$$= a_0 + a_1 x + a_2(a_0) x^2 + a_3(a_1) x^3 + a_4(a_0) x^4 + \cdots$$

$$= \left(a_0 + a_2(a_0) x^2 + a_4(a_0) x^4 + \cdots\right) + \left(a_1 x + a_3(a_1) x^3 + \cdots\right)$$

Legendre Polynomial

Let $(1-x^2)y'' - 2xy' + N(N+1)y = 0$.

Then, $a_{n+2} = \dfrac{n(n+1) - N(N+1)}{(n+1)(n+2)} a_n$, $n = 0, 1, 2, \ldots$

which implies

$a_{N+2} = a_{N+4} = \cdots = 0$; $a_1 = a_3 = \cdots = 0$

Consequently, one of the sol^n is a polynomial of degree N, depending on N is even or odd.

⊛ **Def^n (Legendre Poly^n):-** Let N be a non negative integer. The Legendre Polynomial of degree N, denoted by $P_N(x)$ is defined to be the poly^n sol^n to

$$(1-x^2)y'' - 2xy' + N(N+1)y = 0,$$

which has been normalized so that

$$\boxed{P_N(1) = 1}$$

Example:- $(1-x^2)y'' - 2xy' + 90y = 0.$

$\nu = N = 9$ in the Legendre eq^n.

$$y_1(x) = a_0\left(1 - 45x^2 + 315x^4 - \cdots\cdots\right)$$ (even exponents)

$$y_2(x) = a_1\left(x - 16x^3 + \cdots\cdots + * \; x^9\right) \longrightarrow \boxed{P_9(x)}$$

$\boxed{N=9}$

$*$ $(1-x^2)y'' - 2xy' + 6y = 0$

$\boxed{N=2}$ ✓

$\boxed{P_2(x) = 1 - 3x^2}$

$$y_1(x) = a_0\left(1 - 3x^2 + 0 \cdot x^4 + 0 \cdot x^6 + \cdots\right) = \underbrace{a_0\left(1 - 3x^{\boxed{2}}\right)}_{\text{Poly}^n \text{ of degree } 2}$$

$$y_2(x) = a_1\left(x - \frac{5}{6}x^3 + \cdots\cdots\right) \longrightarrow \text{infinite Series}$$

Power series

$\boxed{N=0}$ $y_1(x) = a_0$ which implies that $P_0(x) = 1$.

$(1-x^2)y'' - 2xy' + y = 0$. — Corresponding Legendre equ^n. (LE)

$\boxed{N=1}$ $y_2(x) = a_1 x$. which implies $P_1(x) = x$.

$(1-x^2)y'' - 2xy' + 2y = 0$ — Corresponding L.E.

$\boxed{N=2}$ $y_1(x) = a_0(1-3x^2)$ and imposing the normalization Cond^n $P_2(1) = 1$, we get $P_2(x) = \frac{1}{2}(3x^2-1)$.

$(1-x^2)y'' - 2xy' + 6y = 0$. — corresponding L.E.

Similarly for $\boxed{N=3}$ the Legendre equation is

$$(1-x^2)y'' - 2xy' + 12y = 0$$

and Legendre polynomial is given by

$$P_3(x) = \frac{1}{2}x(5x^2-3)$$

For $\boxed{N=4}$ the Legendre equation is

$$(1-x^2)y'' - 2xy' + 20y = 0$$

and Legendre polynomial is given by

$$P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3).$$

EXR
Hermite Equation
Chebychev equation

x_0 - as an ordinary point ends here.

x_0 - as a singular point starts here.

Euler-Type Equation:-

$$(x-x_0)^2 y'' + \alpha (x-x_0) y' + \beta y = 0,$$

where α, β are constants.